Android Coursework

Code guide with visual code snapshots

What this covers
The runnable Java Android app in `Android-Project`: splash and startup, fragment navigation, patient list/detail, responsive layouts, issue reporting, notifications, background refresh, and ViewPager2.

Scope

This guide explains implemented project code and shows excerpts from the source files. It excludes theory questions and the separate written answer sheets. Patient records shown in the app are synthetic training data.

Project location: work/Android-Project/app/src/main

1. How the app is organized

The app uses one host activity and a Navigation Component NavHost to switch between feature fragments. Android resources supply screen layouts, menus, navigation destinations, colors, and themes. Several coursework exercises are combined in this runnable project.

app/src/main/AndroidManifest.xml

```
  6         android:label="Android Coursework"
  7         android:supportsRtl="true"
  8         android:theme="@style/Theme.Coursework.Starting"&gt;
  9         &lt;provider
 10             android:name="androidx.startup.InitializationProvider"
 11             android:authorities="${applicationId}.androidx-startup"
 12             android:exported="false"&gt;
 13             &lt;meta-data android:name="com.example.coursework.startup.FirebaseBootstrapInitializer"
 14                 android:value="androidx.startup" /&gt;
 15             &lt;meta-data android:name="com.example.coursework.startup.AnalyticsInitializer"
 16                 android:value="androidx.startup" /&gt;
 17             &lt;meta-data android:name="com.example.coursework.startup.CrashReportingInitializer"
 18                 android:value="androidx.startup" /&gt;
 19         &lt;/provider&gt;
 20         &lt;activity
 21             android:name=".SplashActivity"
 22             android:exported="true"
 23             android:theme="@style/Theme.Coursework.Starting"&gt;
 24             &lt;intent-filter&gt;
 25                 &lt;action android:name="android.intent.action.MAIN" /&gt;
 26                 &lt;category android:name="android.intent.category.LAUNCHER" /&gt;
 27             &lt;/intent-filter&gt;
```

Navigation destinations: app/src/main/res/navigation/main_nav_graph.xml

```
  1 &lt;navigation xmlns:android="http://schemas.android.com/apk/res/android"
  2     xmlns:app="http://schemas.android.com/apk/res-auto" android:id="@+id/main_nav_graph"
  3     app:startDestination="@id/eco_dashboard"&gt;
  4     &lt;fragment android:id="@+id/eco_dashboard" android:name="com.example.coursework.screens.
DashboardFragment" android:label="EcoUganda Dashboard" /&gt;
  5     &lt;fragment android:id="@+id/eco_projects" android:name="com.example.coursework.screens.
ProjectsFragment" android:label="Projects" /&gt;
  6     &lt;fragment android:id="@+id/eco_schools" android:name="com.example.coursework.screens.
SchoolsFragment" android:label="Green Schools" /&gt;
  7     &lt;fragment android:id="@+id/eco_reports" android:name="com.example.coursework.screens.
ReportsFragment" android:label="Reports"&gt;
  8         &lt;argument android:name="reportId" app:argType="string" android:defaultValue="" /&gt;
  9         &lt;deepLink app:uri="ecouganda://reports/{reportId}" /&gt;
 10     &lt;/fragment&gt;
 11     &lt;fragment android:id="@+id/eco_settings" android:name="com.example.coursework.screens.
SettingsFragment" android:label="Settings" /&gt;
 12     &lt;fragment android:id="@+id/uganda_connect" android:name="com.example.coursework.screens.
UgandaConnectFragment" android:label="UgandaConnect" /&gt;
 13     &lt;fragment android:id="@+id/patient_list" android:name="com.example.coursework.patients.
PatientListFragment" android:label="UHC Patient List" /&gt;
 14     &lt;fragment android:id="@+id/patient_detail" android:name="com.example.coursework.patients.
PatientDetailFragment" android:label="Patient Summary"&gt;
 15         &lt;argument android:name="patientId" app:argType="string" /&gt;
 16     &lt;/fragment&gt;
 17     &lt;fragment android:id="@+id/pager_demo" android:name="com.example.coursework.pager.
PagerDemoFragment" android:label="ViewPager2 Demo" /&gt;
 18     &lt;fragment android:id="@+id/report_form" android:name="com.example.coursework.screens.
ReportFormFragment" android:label="Report a service issue" /&gt;
 19 &lt;/navigation&gt;
```

The manifest registers AndroidX App Startup initializers, the launcher splash activity, notification permission, and the EcoUganda report deep link. The navigation graph defines the screens that MainActivity hosts.

2. MainActivity: navigation and app behavior

MainActivity wires the app bar, drawer, phone bottom bar, and tablet navigation rail to a NavController. It also handles toolbar actions, creates the notification channel, and schedules periodic WorkManager work.

app/src/main/java/com/example/coursework/MainActivity.java

```
54            });
55
56    @Override
57    protected void onCreate(Bundle savedInstanceState) {
58        super.onCreate(savedInstanceState);
59        getWindow().setSoftInputMode(WindowManager.LayoutParams.SOFT_INPUT_ADJUST_NOTHING);
60        setContentView(R.layout.activity_main);
61
62        drawerLayout = findViewById(R.id.drawer_layout);
63        toolbar = findViewById(R.id.top_app_bar);
64        drawer = findViewById(R.id.navigation_view);
65        bottomAppBar = findViewById(R.id.bottom_app_bar);
66        fab = findViewById(R.id.main_fab);
67        navigationRail = findViewById(R.id.navigation_rail);
68
69        NavHostFragment host = (NavHostFragment) getSupportFragmentManager()
70                .findFragmentById(R.id.primary_nav_host);
71        if (host == null) throw new IllegalStateException("NavHostFragment missing");
72        navController = host.getNavController();
73        Set&lt;Integer&gt; topLevelDestinations = new HashSet&lt;&gt;(Arrays.asList(
74                R.id.eco_dashboard, R.id.uganda_connect, R.id.patient_list,
75                R.id.eco_projects, R.id.eco_schools, R.id.eco_reports, R.id.eco_settings));
76        AppBarConfiguration appBarConfiguration = new AppBarConfiguration.Builder(topLevelDestinations)
77                .setOpenableLayout(drawerLayout)
78                .build();
79        NavigationUI.setupWithNavController(toolbar, navController, appBarConfiguration);
80        NavigationUI.setupWithNavController(drawer, navController);
81        if (bottomAppBar != null) setupPhoneNavigation();
82        if (navigationRail != null) setupTabletNavigation();
83
84        toolbar.inflateMenu(R.menu.top_app_bar);
85        com.google.android.material.badge.BadgeDrawable badge =
86                com.google.android.material.badge.BadgeDrawable.create(this);
87        badge.setNumber(3);
88        com.google.android.material.badge.BadgeUtils.attachBadgeDrawable(
```

app/src/main/java/com/example/coursework/MainActivity.java

```
95
96    }
97
98    private void setupPhoneNavigation() {
99        Menu menu = bottomAppBar.getMenu();
100       for (int i = 0; i &lt; menu.size(); i++) menu.getItem(i).setCheckable(true);
101       bottomAppBar.setOnMenuItemClickListener(item -&gt; navigateTopLevel(item.getItemId()));
102       navController.addOnDestinationChangedListener((controller, destination, args) -&gt; {
103           MenuItem item = bottomAppBar.getMenu().findItem(destination.getId());
104           if (item != null) item.setChecked(true);
105       });
106   }
107
108   private void setupTabletNavigation() {
109       NavigationUI.setupWithNavController(navigationRail, navController);
110       drawerLayout.setDrawerLockMode(DrawerLayout.LOCK_MODE_LOCKED_OPEN,
111               androidx.core.view.GravityCompat.START);
112   }
113
114   private boolean navigateTopLevel(int destinationId) {
115       if (destinationId == navController.getCurrentDestination().getId()) return true;
116       NavOptions options = new NavOptions.Builder()
117               .setLaunchSingleTop(true)
118               .setRestoreState(true)
119               .setPopUpTo(R.id.eco_dashboard, false, true)
120               .build();
121       navController.navigate(destinationId, null, options);
122       return true;
```

On phones, selecting a bottom bar item navigates to a top-level destination. On tablets, the rail and locked-open drawer layout provide persistent navigation. Window insets hide the bottom controls while the keyboard is visible.

3. Patient list and detail fragments

The patient exercise builds a card list from local synthetic records. Tapping a card selects a patient and either navigates to a detail screen on a phone or updates a detail pane on a tablet.

app/src/main/java/com/example/coursework/patients/PatientListFragment.java

```
17
18 /** Demo UHC patient list with phone detail navigation and tablet two-pane behavior. */
19 public final class PatientListFragment extends Fragment {
20     private static final String[][] PATIENTS = {
21         {"P-1001", "Amina N. · 34 years", "Outpatient follow-up · 12 Sep"},
22         {"P-1002", "Kato M. · 52 years", "Hypertension review · 10 Sep"},
23         {"P-1003", "Grace A. · 27 years", "Antenatal visit · 09 Sep"},
24         {"P-1004", "John O. · 61 years", "Diabetes review · 06 Sep"},
25         {"P-1005", "Sarah K. · 19 years", "Immunization visit · 04 Sep"}
26     };
27     private PatientSelectionViewModel selection;
28     private boolean tabletLayout;
29
30     public PatientListFragment() { super(); }
31
32     @Nullable @Override public View onCreateView(android.view.LayoutInflater inflater,
33             @Nullable android.view.ViewGroup container, @Nullable Bundle state) {
34         selection = new ViewModelProvider(requireActivity()).get(PatientSelectionViewModel.class);
35         boolean tablet = getResources().getConfiguration().smallestScreenWidthDp &gt;= 600;
36         tabletLayout = tablet;
37         LinearLayout root = new LinearLayout(requireContext());
38         root.setOrientation(tablet ? LinearLayout.HORIZONTAL : LinearLayout.VERTICAL);
```

app/src/main/java/com/example/coursework/patients/PatientListFragment.java

```
138         card.addView(row);
139         card.setClickable(true);
140         card.setFocusable(true);
141         card.setOnClickListener(view -&gt; {
142             selection.select(patient[0]);
143             if (getResources().getConfiguration().smallestScreenWidthDp &gt;= 600) {
144                 showTabletDetail(patient[0]);
145             } else {
146                 Bundle args = new Bundle();
147                 args.putString("patientId", patient[0]);
148                 Navigation.findNavController(view).navigate(R.id.patient_detail, args);
149             }
150         });
151         return card;
152     }
153
154     private String getInitials(String nameText) {
155         String[] parts = nameText.split(" ");
156         if (parts.length &gt;= 2) {
157             return (parts[0].substring(0, 1) + parts[1].substring(0, 1)).toUpperCase();
158         }
159         return nameText.substring(0, Math.min(2, nameText.length())).toUpperCase();
```

app/src/main/java/com/example/coursework/patients/PatientSelectionViewModel.java

```
1 package com.example.coursework.patients;
2
3 import androidx.lifecycle.LiveData;
4 import androidx.lifecycle.MutableLiveData;
5 import androidx.lifecycle.ViewModel;
6
7 /** Activity-scoped selection state shared by the patient list and detail fragments. */
8 public final class PatientSelectionViewModel extends ViewModel {
9     private final MutableLiveData&lt;String&gt; selectedPatientId = new MutableLiveData&lt;&gt;();
10     public LiveData&lt;String&gt; getSelectedPatientId() { return selectedPatientId; }
11     public void select(String patientId) { selectedPatientId.setValue(patientId); }
12 }
```

PatientSelectionViewModel is activity-scoped, so fragments can share the selected patient ID. PatientDetailFragment displays a summary and explicitly labels the data as synthetic.

4. Patient detail and responsive presentation

The detail fragment accepts a patient ID in arguments, with the shared ViewModel as a fallback. The tablet list creates a second pane and embeds a child detail fragment, while the phone uses a separate navigation destination.

app/src/main/java/com/example/coursework/patients/PatientDetailFragment.java

```
/** Displays a synthetic patient summary; sample records contain no real patient data. */
public final class PatientDetailFragment extends Fragment {
    public PatientDetailFragment() { super(); }
    public static PatientDetailFragment newInstance(String id) {
        PatientDetailFragment fragment = new PatientDetailFragment();
        Bundle args = new Bundle(); args.putString("patientId", id); fragment.setArguments(args);
        return fragment;
    }

    @Nullable @Override public View onCreateView(android.view.LayoutInflater inflater,
            @Nullable android.view.ViewGroup container, @Nullable Bundle state) {
        String id = getArguments() == null ? null : getArguments().getString("patientId");
        if (id == null &amp;&amp; getActivity() != null) {
            id = new ViewModelProvider(requireActivity()).get(PatientSelectionViewModel.class)
                    .getSelectedPatientId().getValue();
        }

        ScrollView scroll = new ScrollView(requireContext());
        scroll.setFillViewport(true);

        LinearLayout column = new LinearLayout(requireContext());
        column.setOrientation(LinearLayout.VERTICAL);
        int pad = dp(20);
        column.setPadding(pad, pad, pad, dp(110));
```

Tablet resource variant: app/src/main/res/layout-sw600dp/activity_main.xml

```
&lt;androidx.drawerlayout.widget.DrawerLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/drawer_layout" android:layout_width="match_parent" android:
layout_height="match_parent"&gt;
    &lt;androidx.coordinatorlayout.widget.CoordinatorLayout
        android:id="@+id/root_content" android:layout_width="match_parent" android:
layout_height="match_parent"&gt;
        &lt;com.google.android.material.appbar.AppBarLayout android:layout_width="match_parent" android:
layout_height="wrap_content"&gt;
            &lt;com.google.android.material.appbar.MaterialToolbar android:id="@+id/top_app_bar"
                android:layout_width="match_parent" android:layout_height="?attr/actionBarSize"
                android:background="?attr/colorSurface" app:navigationIcon="@drawable/ic_menu"
                app:titleTextColor="?attr/colorOnSurface" /&gt;
        &lt;/com.google.android.material.appbar.AppBarLayout&gt;
        &lt;LinearLayout android:layout_width="match_parent" android:layout_height="match_parent"
            android:layout_marginTop="?attr/actionBarSize" android:orientation="horizontal"&gt;
            &lt;com.google.android.material.navigationrail.NavigationRailView android:id="@+id/
navigation_rail"
                android:layout_width="wrap_content" android:layout_height="match_parent"
                app:menu="@menu/bottom_app_bar" /&gt;
            &lt;androidx.fragment.app.FragmentContainerView android:id="@+id/primary_nav_host"
                android:name="androidx.navigation.fragment.NavHostFragment" android:layout_width="0dp"
```

Android selects the sw600dp resource variant on wider devices. The patient list also checks smallestScreenWidthDp to decide whether to use a side-by-side layout.

5. Service issue report form

ReportFormFragment constructs a Material form with district and issue fields. The submit handler validates that both fields contain text, displays field errors when needed, and shows a local draft confirmation.

app/src/main/java/com/example/coursework/screens/ReportFormFragment.java

```
        form.setOrientation(LinearLayout.VERTICAL);
        form.setPadding(pad, pad, pad, pad);

        TextInputLayout districtLayout = new TextInputLayout(requireContext(), null,
                com.google.android.material.R.attr.textInputOutlinedStyle);
        districtLayout.setHint("District or service area");
        TextInputEditText district = new TextInputEditText(districtLayout.getContext());
        district.setSingleLine(true);
        districtLayout.addView(district);

        TextInputLayout issueLayout = new TextInputLayout(requireContext(), null,
                com.google.android.material.R.attr.textInputOutlinedStyle);
        issueLayout.setHint("Describe the service issue");
        TextInputEditText issue = new TextInputEditText(issueLayout.getContext());
        issue.setMinLines(3);
        issue.setMaxLines(5);
        issueLayout.addView(issue);

        MaterialButton submit = new MaterialButton(requireContext());
        submit.setText("Save report draft");
        LinearLayout.LayoutParams submitParams = new LinearLayout.LayoutParams(-1, -2);
        submitParams.topMargin = dp(20);

        LinearLayout.LayoutParams spacing = new LinearLayout.LayoutParams(-1, -2);
        spacing.topMargin = dp(14);

        form.addView(districtLayout);
        form.addView(issueLayout, spacing);
        form.addView(submit, submitParams);

        card.addView(form);
        page.addView(card);

        submit.setOnClickListener(v -&gt; {
            if (district.getText() == null || district.getText().toString().trim().isEmpty()) {
                districtLayout.setError("Enter a district or service area"); return;
```

app/src/main/java/com/example/coursework/screens/ReportFormFragment.java

```
        card.addView(form);
        page.addView(card);

        submit.setOnClickListener(v -&gt; {
            if (district.getText() == null || district.getText().toString().trim().isEmpty()) {
                districtLayout.setError("Enter a district or service area"); return;
            }
            if (issue.getText() == null || issue.getText().toString().trim().isEmpty()) {
                issueLayout.setError("Describe the issue"); return;
            }
            districtLayout.setError(null); issueLayout.setError(null);
            Snackbar.make(page, "Draft saved on this screen. Connect a service to submit it.",
                    Snackbar.LENGTH_LONG).show();
        });

        scroll.addView(page);
        return scroll;
    }
    private int dp(int n) { return (int) (n * getResources().getDisplayMetrics().density + 0.5f); }
}
```

The current action saves no data and makes no network request. The fragment comment points out that a real service should submit through an authenticated backend.

6. Notification and deep link flow

MainActivity asks for Android 13 notification permission when needed, builds a notification that opens a report URI, and configures a channel on Android 8 and newer. The manifest declares the matching ecouganda://reports/... link.

app/src/main/java/com/example/coursework/MainActivity.java

```
122         return true;
123     }
124 
125     private boolean onToolbarItemSelected(@NonNull MenuItem item) {
126         if (item.getItemId() == R.id.action_search) {
127             navController.navigate(R.id.eco_projects);
128             return true;
129         }
130         if (item.getItemId() == R.id.action_notifications) {
131             if (Build.VERSION.SDK_INT &gt;= 33 &amp;&amp; checkSelfPermission(Manifest.permission.
POST_NOTIFICATIONS)
132                     != android.content.pm.PackageManager.PERMISSION_GRANTED) {
133                 notificationPermission.launch(Manifest.permission.POST_NOTIFICATIONS);
134             } else {
135                 sendReportNotification("UG-2026-001");
136             }
137             return true;
138         }
139         return false;
140     }
141 
142     private void openIssueReport() {
143         if (navController.getCurrentDestination().getId() != R.id.report_form) {
144             navController.navigate(R.id.report_form);
145         }
146     }
147 
148     private void configureResponsiveChrome() {
149         View root = findViewById(R.id.root_content);
150         View appBar = findViewById(R.id.app_bar_layout);
151         ViewCompat.setOnApplyWindowInsetsListener(root, (view, insets) -&gt; {
```

app/src/main/java/com/example/coursework/MainActivity.java

```
172         }
173     }
174 
175     private void sendReportNotification(String reportId) {
176         Uri uri = Uri.parse("ecouganda://reports/" + Uri.encode(reportId));
177         Intent intent = new Intent(Intent.ACTION_VIEW, uri, this, MainActivity.class)
178                 .addFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP | Intent.FLAG_ACTIVITY_SINGLE_TOP);
179         PendingIntent pendingIntent = PendingIntent.getActivity(this, reportId.hashCode(), intent,
180                 PendingIntent.FLAG_UPDATE_CURRENT | PendingIntent.FLAG_IMMUTABLE);
181         android.app.Notification notification = new NotificationCompat.Builder(this, CHANNEL_ID)
182                 .setSmallIcon(R.drawable.ic_layers)
183                 .setContentTitle("EcoUganda report updated")
184                 .setContentText("Tap to view report " + reportId)
185                 .setContentIntent(pendingIntent)
186                 .setAutoCancel(true)
187                 .build();
188         try {
189             NotificationManagerCompat.from(this).notify(reportId.hashCode(), notification);
190         } catch (SecurityException ignored) {
191             // Android 13+: grant POST_NOTIFICATIONS to show notifications.
192         }
193     }
194 
195     private void scheduleOfflineRefresh() {
196         androidx.work.Constraints constraints = new androidx.work.Constraints.Builder()
197                 .setRequiredNetworkType(androidx.work.NetworkType.CONNECTED)
198                 .setRequiresBatteryNotLow(true)
199                 .build();
```

The notification currently references a sample report ID. It is a demonstration of permission handling, channel setup, pending intents, and deep-link navigation.

7. Background refresh hook and app startup

A periodic Worker is enqueued with network and battery constraints. The Worker currently returns success without calling a server, because the project does not include a configured endpoint or credentials.

app/src/main/java/com/example/coursework/MainActivity.java

```
        }
    }

    private void scheduleOfflineRefresh() {
        androidx.work.Constraints constraints = new androidx.work.Constraints.Builder()
                .setRequiredNetworkType(androidx.work.NetworkType.CONNECTED)
                .setRequiresBatteryNotLow(true)
                .build();
        androidx.work.PeriodicWorkRequest request =
                new androidx.work.PeriodicWorkRequest.Builder(
                        com.example.coursework.sync.ProjectRefreshWorker.class,
                        4, java.util.concurrent.TimeUnit.HOURS)
                        .setConstraints(constraints)
                        .build();
        androidx.work.WorkManager.getInstance(this).enqueueUniquePeriodicWork(
                "ecouganda-project-refresh",
                androidx.work.ExistingPeriodicWorkPolicy.KEEP, request);
    }
}
```

app/src/main/java/com/example/coursework/sync/ProjectRefreshWorker.java

```
package com.example.coursework.sync;

import android.content.Context;

import androidx.annotation.NonNull;
import androidx.work.Worker;
import androidx.work.WorkerParameters;

/** Periodic offline refresh hook; replace the sample body with an authenticated repository call. */
public final class ProjectRefreshWorker extends Worker {
    public ProjectRefreshWorker(@NonNull Context context, @NonNull WorkerParameters params) {
        super(context, params);
    }

    @NonNull @Override public Result doWork() {
        if (isStopped()) return Result.retry();
        // The coursework project has no backend endpoint or credentials; this is a safe local demo.
        return Result.success();
    }
}
```

app/src/main/java/com/example/coursework/startup/FirebaseBootstrapInitializer.java

```
package com.example.coursework.startup;

import android.content.Context;

import androidx.annotation.NonNull;
import androidx.startup.Initializer;

import com.google.firebase.FirebaseApp;

import java.util.Collections;
import java.util.List;

/** Initializes Firebase only when the developer has supplied google-services.json. */
public final class FirebaseBootstrapInitializer implements Initializer&lt;Boolean&gt; {
    @NonNull @Override public Boolean create(@NonNull Context context) {
        return FirebaseApp.initializeApp(context) != null;
    }
    @NonNull @Override public List&lt;Class&lt;? extends Initializer&lt;?&gt;&gt;&gt; dependencies() {
        return Collections.emptyList();
    }
}
```

Three Initializer classes are registered by AndroidX App Startup for Firebase bootstrap, analytics, and crash reporting. Service behavior depends on developer-supplied Firebase configuration.

8. ViewPager2 demonstration

PagerDemoFragment places a TabLayout above a ViewPager2. TabLayoutMediator keeps tab labels synchronized with the selected page, while a FragmentStateAdapter creates one independently managed fragment per page.

app/src/main/java/com/example/coursework/pager/PagerDemoFragment.java

```
 14
 15 /** Runnable Task 7 sample using a FragmentStateAdapter and Material tab indicators. */
 16 public final class PagerDemoFragment extends Fragment {
 17     private static final String[] TITLES = {"Welcome", "Services", "Offline"};
 18     public PagerDemoFragment() { super(); }
 19
 20     @Nullable @Override public View onCreateView(android.view.LayoutInflater inflater,
 21             @Nullable ViewGroup container, @Nullable Bundle state) {
 22         LinearLayout layout = new LinearLayout(requireContext());
 23         layout.setOrientation(LinearLayout.VERTICAL);
 24         TabLayout tabs = new TabLayout(requireContext());
 25         ViewPager2 pager = new ViewPager2(requireContext());
 26         pager.setId(View.generateViewId());
 27         layout.addView(tabs, new LinearLayout.LayoutParams(-1, -2));
 28         layout.addView(pager, new LinearLayout.LayoutParams(-1, 0, 1));
 29         pager.setAdapter(new PageAdapter(this));
 30         new TabLayoutMediator(tabs, pager, (tab, position) -&gt; tab.setText(TITLES[position])).attach(
);
 31         return layout;
 32     }
 33
 34     private static final class PageAdapter extends androidx.viewpager2.adapter.FragmentStateAdapter {
 35         PageAdapter(Fragment host) { super(host); }
 36         @Override public int getItemCount() { return TITLES.length; }
 37         @Override public Fragment createFragment(int position) {
 38             return PagerPageFragment.newInstance(TITLES[position],
 39                     "Swipe between pages. Each page is an independently managed Fragment.");
 40         }
 41     }
 42 }
```

app/src/main/java/com/example/coursework/pager/PagerPageFragment.java

```
  1 package com.example.coursework.pager;
  2
  3 import android.os.Bundle;
  4 import android.view.View;
  5 import android.view.ViewGroup;
  6 import android.widget.TextView;
  7
  8 import androidx.annotation.Nullable;
  9 import androidx.fragment.app.Fragment;
 10
 11 public final class PagerPageFragment extends Fragment {
 12     public PagerPageFragment() { super(); }
 13     static PagerPageFragment newInstance(String title, String body) {
 14         PagerPageFragment fragment = new PagerPageFragment();
 15         Bundle args = new Bundle(); args.putString("title", title); args.putString("body", body);
 16         fragment.setArguments(args); return fragment;
 17     }
 18     @Nullable @Override public View onCreateView(android.view.LayoutInflater inflater,
 19             @Nullable ViewGroup container, @Nullable Bundle state) {
 20         TextView page = new TextView(requireContext());
 21         Bundle args = requireArguments();
 22         page.setText(args.getString("title") + "\n\n" + args.getString("body"));
 23         page.setTextSize(22); page.setGravity(android.view.Gravity.CENTER);
 24         int pad = (int) (24 * getResources().getDisplayMetrics().density);
 25         page.setPadding(pad, pad, pad, pad); return page;
 26     }
 27 }
```

This is the runnable code example for the ViewPager2 task; the theory assignment folders and their written questions are intentionally omitted.

9. UgandaConnect and EcoUganda screens

The project includes feature fragments for the UgandaConnect service navigation and EcoUganda dashboard, projects, schools, reports, settings, and card screens. The navigation graph and menus connect these destinations to the shared activity chrome.

app/src/main/java/com/example/coursework/fragments/NavigationOneFragment.java

```
 16 public final class NavigationOneFragment extends Fragment {
 17     @Nullable
 18     @Override
 19     public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup container,
 20                              @Nullable Bundle savedInstanceState) {
 21         return inflater.inflate(R.layout.fragment_navigation_one, container, false);
 22     }
 23
 24     @Override
 25     public void onViewCreated(@NonNull View view, @Nullable Bundle savedInstanceState) {
 26         view.findViewById(R.id.action_primary).setOnClickListener(
 27                 ignored -&gt; Snackbar.make(view, R.string.action_completed, Snackbar.LENGTH_SHORT).
show());
 28     }
 29 }
```

app/src/main/java/com/example/coursework/fragments/NavigationTwoFragment.java

```
 15 public final class NavigationTwoFragment extends Fragment {
 16     @Nullable
 17     @Override
 18     public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup container,
 19                              @Nullable Bundle savedInstanceState) {
 20         return inflater.inflate(R.layout.fragment_navigation_two, container, false);
 21     }
 22 }
```

app/src/main/java/com/example/coursework/screens/UgandaConnectFragment.java

```
  1 package com.example.coursework.screens;
  2
  3 public final class UgandaConnectFragment extends CardScreenFragment {
  4     @Override protected String heading() { return "UgandaConnect services"; }
  5     @Override protected String[] entries() { return new String[] {
  6         "Solar Subsidies\nREA application status and community installations",
  7         "My Results\nUNEB examination results and term dates",
  8         "Hospital Services\nAppointments and results",
  9         "Bill Payments\nUMEME and NWSC account status",
 10         "KCCA Reports\nReport a city service issue",
 11         "Bookmarks\nSave frequently used public services"
 12     }; }
 13 }
```

app/src/main/java/com/example/coursework/screens/CardScreenFragment.java

```
 17     protected abstract String heading();
 18     protected abstract String[] entries();
 19
 20     @Nullable @Override public android.view.View onCreateView(android.view.LayoutInflater inflater,
 21             @Nullable ViewGroup container, @Nullable Bundle savedInstanceState) {
 22         ScrollView scroll = new ScrollView(requireContext());
 23         scroll.setFillViewport(true);
 24         LinearLayout column = new LinearLayout(requireContext());
 25         column.setOrientation(LinearLayout.VERTICAL);
 26         int pad = dp(20);
 27         column.setPadding(pad, pad, pad, dp(110));
 28
 29         if (heading() != null &amp;&amp; !heading().trim().isEmpty()) {
 30             MaterialTextView title = new MaterialTextView(requireContext());
 31             title.setText(heading());
 32             title.setTextAppearance(com.google.android.material.R.style.
TextAppearance_Material3_HeadlineSmall);
```

The smaller dashboard and destination fragments are mostly lightweight screen stubs. Supporting menu and layout XML files define their navigation labels and app bar presentation.

10. Build and project notes

Open the Android-Project directory in Android Studio. The repository README identifies it as the single buildable project and maps coursework tasks to runnable examples. Firebase and map/network features need developer configuration before they can use real services.

App dependencies and Android configuration: app/build.gradle.kts

```
plugins {
    id("com.android.application")
}

android {
    namespace = "com.example.coursework"
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.coursework"
        minSdk = 23
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_17
        targetCompatibility = JavaVersion.VERSION_17
    }
}

dependencies {
    implementation("androidx.appcompat:appcompat:1.7.0")
    implementation("androidx.core:core:1.15.0")
    implementation("androidx.core:core-splashscreen:1.2.0")
    implementation("androidx.drawerlayout:drawerlayout:1.2.0")
    implementation("androidx.fragment:fragment:1.8.5")
    implementation("androidx.navigation:navigation-fragment:2.8.5")
    implementation("androidx.navigation:navigation-ui:2.8.5")
    implementation("androidx.lifecycle:lifecycle-viewmodel:2.8.7")
    implementation("androidx.lifecycle:lifecycle-livedata:2.8.7")
```

app/src/main/res/values/strings.xml

```
&lt;resources&gt;
    &lt;string name="app_name"&gt;Android Coursework&lt;/string&gt;
    &lt;string name="title_home"&gt;Home&lt;/string&gt;
    &lt;string name="title_fragments"&gt;Fragments&lt;/string&gt;
    &lt;string name="title_navigation_one"&gt;Navigation One&lt;/string&gt;
    &lt;string name="title_navigation_two"&gt;Navigation Two&lt;/string&gt;
    &lt;string name="title_about"&gt;About&lt;/string&gt;
    &lt;string name="drawer_header_title"&gt;Android Coursework&lt;/string&gt;
    &lt;string name="drawer_header_subtitle"&gt;Fragments and Material navigation&lt;/string&gt;
    &lt;string name="home_heading"&gt;Coursework demonstrations&lt;/string&gt;
    &lt;string name="home_body"&gt;Use the app bar menu to open Navigation One. Open the navigation
drawer to reach Navigation Two and the fragment demonstration. The app starts with the Android SplashScreen
API.&lt;/string&gt;
    &lt;string name="fragments_heading"&gt;Task 2: Android Fragments&lt;/string&gt;
    &lt;string name="first_fragment"&gt;First fragment&lt;/string&gt;
    &lt;string name="second_fragment"&gt;Second fragment&lt;/string&gt;
    &lt;string name="first_content"&gt;First content fragment&lt;/string&gt;
    &lt;string name="second_content"&gt;Second content fragment&lt;/string&gt;
    &lt;string name="navigation_one_heading"&gt;Task 5: App bar navigation&lt;/string&gt;
    &lt;string name="navigation_one_body"&gt;This destination is opened from the Material app bar
options menu. The app bar remains available while the fragment content changes.&lt;/string&gt;
    &lt;string name="action_primary"&gt;Run action&lt;/string&gt;
    &lt;string name="action_completed"&gt;Action completed&lt;/string&gt;
    &lt;string name="navigation_two_heading"&gt;Task 6: Navigation drawer&lt;/string&gt;
    &lt;string name="navigation_two_body"&gt;This destination is reached from the navigation drawer.
The drawer uses a single checked destination group and closes after a selection.&lt;/string&gt;
```

Included code areas

Splash activity and AndroidX startup hooks; Material navigation and adaptive layouts; UHC patient list/detail; report form; local notification/deep link; WorkManager refresh hook; ViewPager2 fragment adapter. Theory question sheets were excluded.